

Exercise JA

Q1. The slope of the tangent to the curve $(y - x^5)^2 = x(1+x^2)^2$ at the point (1,3) is

Ans. 8

Sol. $(y - x^5)^2 = x(1 + x^2)^2$

$$2(y - x^5) \left(\frac{dy}{dx} - 5x^4 \right)$$

$$= (1 + x^2)^2 + 2x(1 + x^2) 2x$$

at point (1, 3)

$$\therefore 2(3 - 1) \left(\frac{dy}{dx} - 5 \right) = 4 + 8$$

$$\frac{dy}{dx} - 5 = \frac{12}{4} = 3$$

$$\frac{dy}{dx} = 8$$

Paragraph

Let $f(x) = (1 - x)^2 \sin^2 x + x^2$ for all $x \in \mathbb{R}$ and let $g(x) = \int_1^x \left(\frac{2(t-1)}{t+1} - \ln t \right)$ for all $x \in (1, \infty)$.

\n

Q2. Which of the following is true ?

(A) g is increasing on $(1, \infty)$ (B) g is decreasing on $(1, \infty)$

(C) g is increasing on $(1, 2)$ and decreasing on $(2, \infty)$

(D) g is decreasing on $(1, 2)$ and increasing on $(2, \infty)$

Ans. B

Sol. $f(x) = (1 - x)^2 \sin^2 x + x^2$

P : $f(x) + 2x = 2(1 + x^2)$

$$\Rightarrow (1 - x)^2 \sin^2 x + x^2 + 2x = 2 + 2x^2$$

$$\Rightarrow (1 - x)^2 \sin^2 x - x^2 + 2x - 2 = 0$$

$$(1 - x)^2 \cos^2 x + 1 = 0$$

which is not possible.

$\therefore$ P is false.

Q : $2f(x) + 1 = 2x(1 + x)$

$$2x^2 + 2(1 - x)^2 \sin^2 x + 1 = 2x^2 + 2x$$

$$2(1 - x)^2 \sin^2 x - 2x + 1 = 0.$$

Let $h(x) = 2(1 - x)^2 \sin^2 x - 2x + 1$,

clearly $h(1) = -1$ and $h(x) = 2(x^2 - 2x + 1)\sin^2 x - 2x + 1$

$$= x^2 \left[2 \left(1 - \frac{2}{x} + \frac{1}{x^2} \right) \cdot \sin^2 x - \frac{2}{x} + \frac{1}{x^2} \right]$$

$\therefore h(x) \rightarrow \infty$ as $x \rightarrow \infty$.

$\therefore$ By intermediate value theorem $h(x) = 0$ has a root which is greater than 1.

Hence Q is true.

Paragraph

Let $f(x) = (1 - x)^2 \sin^2 x + x^2$ for all $x \in \mathbb{R}$ and let $g(x) = \int_1^x \left(\frac{2(t-1)}{t+1} - \ln t \right)$ for all $x \in (1, \infty)$.

\n

Q3. Consider the statements :

P : There exists some $x \in \mathbb{R}$ such that $f(x) + 2x = 2(1 + x^2)$

Q : There exists some $x \in \mathbb{R}$ such that $2f(x) + 1 = 2x(1 + x)$

Then

(A) both P and Q are true (B) P is true and Q are false (C) P is false and Q is true

(D) both P and Q are false

Ans. C

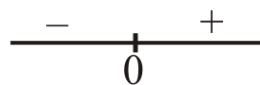
Sol. -

Q4. The number of points in $(-\infty, \infty)$, for which $x^2 - x \sin x - \cos x = 0$, is :

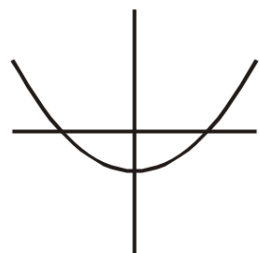
(A) 6 (B) 4 (C) 2 (D) 0

Ans. C

Sol. $f(x) = x^2 - x \sin x - \cos x$
 $f'(x) = 2x - x \cos x - \sin x + \sin x$
 $= x(2 - \cos x)$



$\therefore$ graph of $f(x)$ will be



$\therefore f(x)$ is zero for 2 values of x

$\therefore$ (c)

- Q5. For every pair of continuous functions $f, g : [0,1] \rightarrow \mathbb{R}$ such that $\max \{f(x) : x \in [0,1]\} = \max \{g(x) : x \in [0,1]\}$, the correct statement(s) is(are) :
- (A) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0,1]$
- (B) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0,1]$
- (C) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$ for some $c \in [0,1]$
- (D) $(f(c))^2 = (g(c))^2$ for some $c \in [0,1]$

Ans. A, D

Sol. Consider

$$h(x) = f(x) - g(x) \quad \text{Assume } a < b$$

$$h(a) = \lambda - g(a) > 0$$

$$h(b) = f(b) - \lambda < 0$$

else if $a > b$ $h(a) < 0$ and $h(b) > 0$.

By intermediate value theorem $\Rightarrow h(c) = 0$ (1)

$$\begin{aligned} \text{(A)} \quad & (f(c))^2 + 3f(c) = (g(c))^2 + 3g(c) \\ & (f(c) - g(c))(f(c) + g(c) + 3) = 0 \\ & \text{So there exist a 'c' : } f(c) - g(c) \\ & \text{from (1).} \end{aligned}$$

Hence A is correct.

$$\begin{aligned} \text{(D)} \quad & \text{Similarly } (f(c))^2 = (g(c))^2 \\ & (f(c) - g(c))(f(c) + g(c)) = 0 \\ & \Rightarrow \text{(D) is correct.} \end{aligned}$$

B & C are wrong as by counter eg

If $f(x) = g(x) = \lambda \neq 0$, then

$B \rightarrow \lambda^2 + \lambda = \lambda^2 + 3\lambda$ is not possible.

$C \rightarrow \lambda^2 + 3\lambda = \lambda^2 + \lambda$ is not possible.

- Q6. Let $f, g : [-1, 2] \rightarrow \mathbb{R}$ be continuous functions which are twice differentiable on the interval $(-1, 2)$. Let the values of f and g at the points $-1, 0$ and 2 be as given in the following table :

	$x = -1$	$x = 0$	$x = 2$
$f(x)$	3	6	0
$g(x)$	0	1	-1

In each of the intervals $(-1, 0)$ and $(0, 2)$ the function $(f - 3g)''$ never vanishes. Then the correct statement(s) is (are)

- (A) $f'(x) - 3g'(x) = 0$ has exactly three solutions in $(-1, 0) \cup (0, 2)$
- (B) $f'(x) - 3g'(x) = 0$ has exactly one solutions in $(-1, 0)$

(C) $f(x) - 3g'(x) = 0$ has exactly one solutions in $(0,2)$

(D) $f(x) - 3g'(x) = 0$ has exactly two solutions in $(-1, 0)$ and exactly two solutions in $(0,2)$

Ans. C, B

Sol. -

Paragraph

matching the information given in the three columns of the following table.

Let $f(x) = x + \log_e x - x \log_e x$, $x \in (0, \infty)$

Column 1 contains information about zeros of $f(x)$, $f'(x)$ and $f''(x)$.

Column 2 contains information about the limiting behavior of $f(x)$, $f'(x)$ and $f''(x)$ at infinity.

Column 3 contains information about the increasing/decreasing nature of $f(x)$, $f'(x)$

	Column-1	Column-2	Column-3
(I)	$f(x) = 0$ for some $x \in (1, e^2)$	(i) $\lim_{x \rightarrow \infty} f(x) = 0$	(P) f is increasing in $(0, 1)$
(II)	$f'(x) = 0$ for some $x \in (1, e)$	(ii) $\lim_{x \rightarrow \infty} f(x) = -\infty$	(Q) f is decreasing in (e, e^2)
(III)	$f''(x) = 0$ for some $x \in (0, 1)$	(iii) $\lim_{x \rightarrow \infty} f'(x) = -\infty$	(R) f' is increasing in $(0, 1)$
(IV)	$f''(x) = 0$ for some $x \in (1, e)$	(iv) $\lim_{x \rightarrow \infty} f''(x) = 0$	(S) f' is decreasing in (e, e^2)

Q7. Which of the following options is the only CORRECT combination ?

(A) (II) (ii) (Q) (B) (III) (iii) (R) (C) (IV) (iv) (S) (D) (I) (i) (P)

Ans. A

Sol. -

Paragraph

matching the information given in the three columns of the following table.

Let $f(x) = x + \log_e x - x \log_e x$, $x \in (0, \infty)$

Column 1 contains information about zeros of $f(x)$, $f'(x)$ and $f''(x)$.

Column 2 contains information about the limiting behavior of $f(x)$, $f'(x)$ and $f''(x)$ at infinity.

Column 3 contains information about the increasing/decreasing nature of $f(x)$, $f'(x)$

	Column-1	Column-2	Column-3
(I)	$f(x) = 0$ for some $x \in (1, e^2)$	(i) $\lim_{x \rightarrow \infty} f(x) = 0$	(P) f is increasing in $(0, 1)$

- (II) $f(x) = 0$ for some $x \in (1, e)$ (ii) $\lim_{x \rightarrow \infty} f(x) = -\infty$ (Q) f is decreasing in (e, e^2)
- (III) $f(x) = 0$ for some $x \in (0, 1)$ (iii) $\lim_{x \rightarrow \infty} f(x) = -\infty$ (R) f is increasing in $(0, 1)$
- (IV) $f'(x) = 0$ for some $x \in (1, e)$ (iv) $\lim_{x \rightarrow \infty} f'(x) = 0$ (S) f' is decreasing in (e, e^2)

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Q8. Which of the following options is the only CORRECT combination?

- (A) (III) (iv) (P) (B) (II) (iii) (S) (C) (I) (ii) (R) (D) (IV) (i) (S)

Ans. B

Sol. -

Paragraph

matching the information given in the three columns of the following table.

\nLet $f(x) = x + \log_e x - x \log_e x$, $x \in (0, \infty)$

\nColumn 1 contains information about zeros of $f(x)$, $f'(x)$ and $f''(x)$.

\nColumn 2 contains information about the limiting behavior of $f(x)$, $f'(x)$ and $f''(x)$ at infinity.

\nColumn 3 contains information about the increasing/decreasing nature of $f(x)$, $f'(x)$

\n	Column-1	Column-2	Column-3
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\n(I) $f(x) = 0$ for some $x \in (1, e^2)$	(i) $\lim_{x \rightarrow \infty} f(x) = 0$	(P) f is increasing in $(0, 1)$
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\n(II) $f'(x) = 0$ for some $x \in (1, e)$	(ii) $\lim_{x \rightarrow \infty} f(x) = -\infty$	(Q) f is decreasing in (e, e^2)
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\n(III) $f(x) = 0$ for some $x \in (0, 1)$	(iii) $\lim_{x \rightarrow \infty} f'(x) = -\infty$	(R) f' is increasing in $(0, 1)$
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\n(IV) $f''(x) = 0$ for some $x \in (1, e)$	(iv) $\lim_{x \rightarrow \infty} f''(x) = 0$	(S) f' is decreasing in (e, e^2)
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Q9. Which of the following option is the only INCORRECT combination?

- (A) (I) (iii) (P) (B) (II) (iii) (P) (C) (III) (i) (R) (D) (II) (iv) (Q)

Ans. C

Sol. -

Q10. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a twice differentiable function such that $f'(x) > 0$ for all $x \in \mathbb{R}$, and $f\left(\frac{1}{2}\right) = \frac{1}{2}, f(1) = 1$ then:

- (A) $\frac{1}{2} < f'(1) \leq 1$ (B) $0 < f'(1) \leq \frac{1}{2}$ (C) $f'(1) \leq 0$ (D) $f'(1) > 1$

Ans. D

Sol. $f'(x) > 0$ for all $x \in \mathbb{R}$, $f(1/2) = 1/2$, $f(1) = 1$

$\Rightarrow f'(x)$ increases

Let $g(x) = f(x) - x$, $x \in [1/2, 1)$

$f'(x) = 1$ has at least one real root in $(1/2, 1)$

Hence $f'(x)$ increases $\Rightarrow f'(1) > 1$

Q11. For every twice differentiable function $f: \mathbb{R} \rightarrow [-2, 2]$ with $(f(0))^2 + (f'(0))^2 = 85$, which of the following statement(s) is (are) TRUE ?

(A) There exist $r, s \in \mathbb{R}$, where $r < s$, such that f is one-one on the open interval (r, s)

(B) There exists $x_0 \in (-4, 0)$ such that $|f(x_0)| \leq 1$ (C) $\lim_{x \rightarrow \infty} f(x) = 1$

(D) There exists $\alpha \in (-4, 4)$ such that $f(\alpha) + f'(\alpha) = 0$ and $f'(\alpha) \neq 0$

Ans. A, B, D

Sol. -

Q12. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$f(x) = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$ Then which of the following statements is (are) TRUE ?

(A) f is decreasing in the interval $(-2, -1)$ (B) f is increasing in the interval $(1, 2)$

(C) f is onto (D) Range of f is $\left[-\frac{3}{2}, 2\right]$

Ans. A, B

Sol. $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$$

$$f'(x) = \frac{(2x - 3)(x^2 + 2x + 4) - (2x + 2)(x^2 - 3x - 6)}{(x^2 + 2x + 4)^2}$$

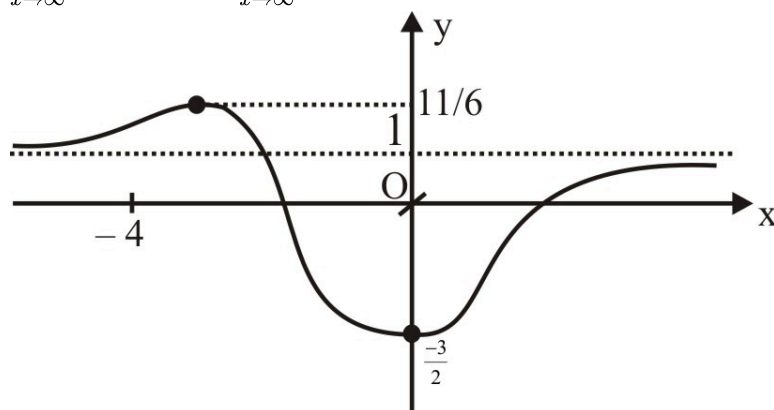
$$f'(x) = \frac{5x(x + 4)}{(x^2 + 2x + 4)^2} \Rightarrow \begin{array}{c} + \quad - \\ | \quad | \\ -4 \quad 0 \end{array}$$

$f(x)$ is decreasing in $(-2, -1) \Rightarrow$ (A) is correct

$f(x)$ is increasing in $(1, 2) \Rightarrow$ (B) is correct

$$\text{Now, } f(-4) = \frac{16 + 12 - 6}{16 - 8 + 4} = \frac{11}{6}; f(0) = \frac{-3}{2}$$

$$\lim_{x \rightarrow -\infty} f(x) = 1 \text{ \& } \lim_{x \rightarrow \infty} f(x) = 1$$



Range is $\left[-\frac{3}{2}, \frac{11}{6}\right] \Rightarrow$ (D) is incorrect

Range $\neq$ Co-domain $\Rightarrow$ (C) is incorrect

Q13. Let S be the set of all twice differentiable functions f from $\mathbb{R}$ to $\mathbb{R}$ such that $\frac{d^2 f}{dx^2}(x) > 0$ for all $x \in (-1, 1)$. For $f \in S$, let X_f be the number of points $x \in (-1, 1)$ for which $f(x) = x$. Then which of the following statements is (are) true?

- (A) There exists a function $f \in S$ such that $X_f = 0$
- (B) For every function $f \in S$, we have $X_f \leq 2$
- (C) There exists a function $f \in S$ such that $X_f = 2$
- (D) There does NOT exist any function f in such that $X_f = 1$

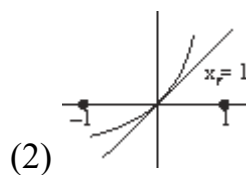
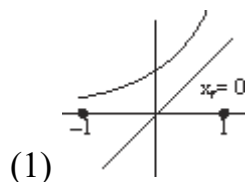
Ans. A, B, C

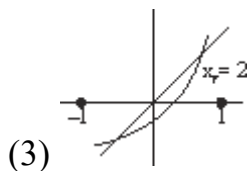
Sol. S = set of all twice differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$

$$\frac{d^2 f}{dx^2} > 0 \text{ in } (-1, 1)$$

Graph 'f' is Concave upward

Number of solutions of $f(x) = x \rightarrow x_f$





$\Rightarrow$ Graph of $y = f(x)$ can intersect graph of $y = x$ at atmost two points $\Rightarrow 0 \leq x_f \leq 2$

Aliter

$$\frac{d^2 f(x)}{dx^2} > 0$$

Let $\phi(x) = f(x) - x$

$$\phi''(x) > 0$$

$\therefore \phi'(x) > 0$ has atmost 1 root in $x \in (-1, 1)$

$\therefore \phi(x) > 0$ has atmost 2 root in $x \in (-1, 1)$

$\therefore x_f \leq 2$

Q14. If the function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$f(x) = \frac{\sin x}{e^{\pi x}} \frac{(x^{2023} + 2024x + 2025)}{(x^2 - x + 3)} + \frac{2}{e^{\pi x}} \frac{(x^{2023} + 2024x + 2025)}{(x^2 - x + 3)}$, then find out the number of solutions of $f(x) = 0$ in $\mathbb{R}$.

Ans. 1

Sol. $f(x) = \frac{\sin x}{e^{\pi x}} \frac{(x^{2023} + 2024x + 2025)}{(x^2 - x + 3)} + \frac{2}{e^{\pi x}} \frac{(x^{2023} + 2024x + 2025)}{(x^2 - x + 3)}$

$$\Rightarrow \frac{1}{e^{\pi x}} \frac{(x^{2023} + 2024x + 2025)}{(x^2 - x + 3)} [\sin x + 2] = 0$$

Now $\sin x + 2 = 0$

$\sin x = -2$ (No solution)

So now, we need to solve

$$x^{2023} + 2024x + 2025 = 0$$

Let $g(x) = x^{2023} + 2024x + 2025$

Differentiate

$$g'(x) = 2023x^{2022} + 2024 \text{ (Always positive)}$$

So It cuts x -axis only once.

So exactly 1 solution.

Q15. If $f(x) = \int_0^x e^{t^2} (t-2)(t-3) dt$ for all $x \in (0, \infty)$, then –

(A) f has a local maximum at $x = 2$ (B) f is decreasing on $(2, 3)$

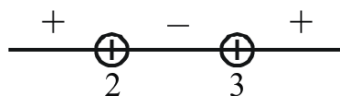
(C) there exists some $c \in (0, \infty)$ such that $f'(c) = 0$

(D) f has a local minimum at $x = 3$

Ans. A, B, C, D

Sol.

$$f(x) = \int_0^x e^{t^2} (t-2)(t-3) dt$$



$$\Rightarrow f'(x) = e^{x^2} (x-2)(x-3)$$

$$\therefore f'(2) = f'(3) = 0$$

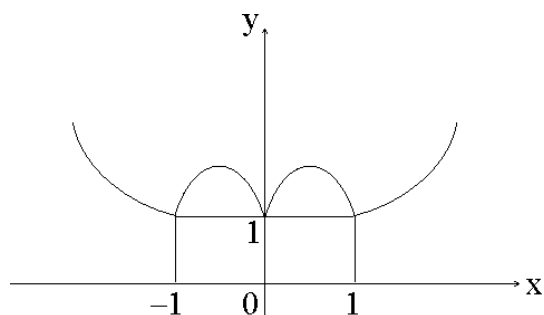
$$\Rightarrow f''(c) = 0 \text{ for some } c \in (2, 3) \text{ (by Rolle's theorem)}$$

Q16. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = |x| + |x^2 - 1|$. The total number of points at which f attains either a local maximum or a local minimum is :

Ans. (a) 5, (b) 9

Sol. $f(x) = |x| + |(x+1)(x-1)|$

$$\Rightarrow f(x) = \begin{cases} x^2 - x - 1 & x < -1 \\ -x^2 - x + 1 & -1 \leq x < 0 \\ -x^2 + x + 1 & 0 \leq x < 1 \\ x^2 + x - 1 & x \geq 1 \end{cases}$$



$\therefore f$ has 5 points where it attains either a local maximum or local minimum.

Q17. Let $p(x)$ be a real polynomial of least degree which has a local maximum at $x = 1$ and a local minimum at $x = 3$. If $p(1) = 6$ and $p(3) = 2$, then $p'(0)$ is

Ans. 9

Sol. -

Q18. The function $f(x) = 2|x| + |x+2| - ||x+2|-2||x||$ has a local minimum or a local maximum at $x =$

- (A) -2 (B) $-\frac{2}{3}$ (C) 2 (D) $\frac{2}{3}$

Ans. A

Sol. -

Paragraph

Let $f : (0, 1) \rightarrow \mathbb{R}$ (the set of all real numbers) be a function. Suppose the function f is twice differentiable, $f(0) = f(1) = 0$ and satisfies $f''(x) - 2f'(x) + f(x) \geq e^x$, $x \in [0, 1]$.

\n

Q19. Which of the following is true for $0 < x < 1$?

- (A) $0 < f(x) < \infty$ (B) $-\frac{1}{2} < f(x) < \frac{1}{2}$ (C) $-\frac{1}{4} < f(x) < 1$ (D) $-\infty < f(x) < 1$

Ans. D

Sol. -

Q20. If the function $e^{-x} f(x)$ assumes its minimum in the interval $[0, 1]$ at $x = \frac{1}{4}$, which of the following is true ?

- (A) $f'(x) < f(x)$, $\frac{1}{4} < x < \frac{3}{4}$ (B) $f'(x) > f(x)$, $0 < x < \frac{1}{4}$ (C) $f'(x) < f(x)$, $0 < x < \frac{1}{4}$
(D) $f'(x) < f(x)$, $\frac{3}{4} < x < 1$

Ans. C

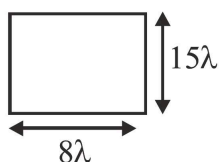
Sol. -

Q21. A rectangular sheet of fixed perimeter with sides having their lengths in the ratio 8 : 15 is converted into an open rectangular box by folding after removing squares of equal area from all four corners. If the total area of removed squares is 100, the resulting box has maximum volume. Then the lengths of the sides of the rectangular sheet are :

- (A) 24 (B) 32 (C) 45 (D) 60

Ans. A, C

Sol. Where $P = 8\lambda + 15\lambda + 8\lambda + 15\lambda$ & λ is constant



Let removed length from each sides is x

Removed area is $4x^2 = 100 \Rightarrow x = 5$

$$V = (8\lambda - 2x)(15\lambda - 2x)x$$

$$V = 120\lambda^2x - 46\lambda x^2 + 4x^3$$

$$\frac{dv}{dx} = 120\lambda^2 - 92\lambda x + 12x^2 = 0$$

$$\text{Put } x = 5 \Rightarrow 120\lambda^2 - 460\lambda + 300 = 0$$

$$12\lambda^2 - 46\lambda + 30 = 0$$

$$6\lambda^2 - 23\lambda + 15 = 0$$

$$(\lambda - 3)(6\lambda - 5) = 0$$

$$\lambda = 3 \text{ \& } \lambda = \frac{5}{6}$$

$$\frac{d^2v}{dx^2} = -92\lambda + 24x = 120 - 92\lambda$$

$$\text{at } \lambda = 3 \Rightarrow \frac{d^2v}{dx^2} < 0$$

$$\text{at } \lambda = \frac{5}{6} \Rightarrow \frac{d^2v}{dx^2} > 0 \text{ (rejected)}$$

Q22. Let $a \in \mathbb{R}$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = x^5 - 5x + a$. Then

(A) $f(x)$ has three real roots if $a > 4$ (B) $f(x)$ has only one real root if $a > 4$

(C) $f(x)$ has three real roots if $a < -4$ (D) $f(x)$ has three real roots if $-4 < a < 4$

Ans. B, D

$$\text{Sol. } f(x) = x^5 - 5x + a = 0$$

$$x^5 - 5x = -a$$

$$x(x^4 - 5) = -a$$

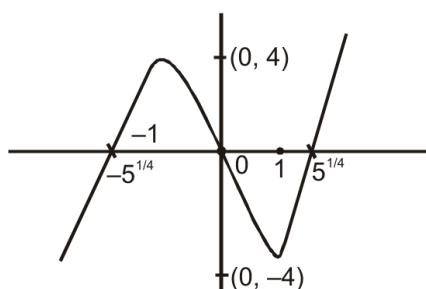
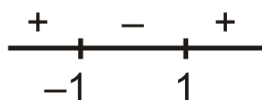
$$x(x^2 - \sqrt{5})(x^2 + \sqrt{5}) = -a$$

$$x(x - 5^{1/4})(x + 5^{1/4})(x^2 + \sqrt{5}) = -a \quad \dots\dots(1)$$

$$f(x) = 5x^4 - 5 = 0$$

$$(x^2 - 1)(x^2 + 1) = 0$$

$$(x - 1)(x + 1)(x^2 + 1) = 0$$



Q23. A cylindrical container is to be made from certain solid material with the following constraints. It has a fixed inner volume of $V \text{ mm}^3$, has a 2 mm thick wall and is open at the top. The bottom of the container is a solid circular disc of thickness 2 mm and is of radius equal to the outer radius of the container.

If the volume of the material used to make the container is minimum when the inner radius of the container is 10 mm, then the value of $\frac{V}{250\pi}$ is :

Ans. 4

Sol. Volume of material $V = \pi r^2 h$

$$\Rightarrow V_1 = \pi(r+2)^2 \cdot 2 + \pi(r+2)^2 h - \pi r^2 h$$

$$\Rightarrow V_1 = 2\pi(r+2)^2 + \pi h(4+4r)$$

$$\Rightarrow V_1 = 2\pi(r+2)^2 + 4\pi h(r+1)$$

$$\Rightarrow V_1 = 2\pi \left((r+2)^2 + \frac{2(r+1)V}{\pi r^2} \right)$$

$$\Rightarrow \frac{dV_1}{dr} = 2\pi \left(2(r+2) + \frac{2v}{\pi} \left(\frac{-1}{r^2} - \frac{2}{r^3} \right) \right) = 0$$

$$\Rightarrow 24 + \frac{2V}{\pi} \left(\frac{-2-10}{10^3} \right) = 0$$

$$\Rightarrow \frac{24v}{10^3\pi} = 24 \Rightarrow v = 10^3\pi$$

$$\Rightarrow \frac{v}{250\pi} = 4$$

Q24. Let $f: \square \rightarrow (0, \infty)$ and $f: \square \rightarrow \square$ be twice differentiable functions such that f' and g'' are continuous functions of $\square$. Suppose $f'(2) = g(2) = 0$, $f''(2) \neq 0$ and $g'(2) \neq 0$. If $\lim_{x \rightarrow 2} \frac{f(x)g(x)}{f'(x)g'(x)} = 1$, then

(A) f has a local minimum at $x = 2$ (B) f has a local maximum at $x = 2$

(C) $f''(2) > f(2)$ (D) $f(x) - f''(x) = 0$ for at least one $x \in \square$

Ans. A, D

Sol. -

Q25. If $f(x) = \begin{vmatrix} \cos(2x) & \cos(2x) & \sin(2x) \\ -\cos x & \cos x & -\sin x \\ \sin x & \sin x & \cos x \end{vmatrix}$, then :

(A) $f(x)$ attains its maximum at $x = 0$ (B) $f(x)$ attains its minimum at $x = 0$

(C) $f'(x) = 0$ at more than three points in $(-\pi, \pi)$

(D) $f'(x) = 0$ at exactly three points in $(-\pi, \pi)$

Ans. A, C

Sol. -

Q26. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f(x) = \begin{cases} x^5 + 5x^4 + 10x^3 + 10x^2 + 3x + 1 & , \quad x < 0; \\ x^2 - x + 1 & , \quad 0 \leq x < 1; \\ \frac{2}{3}x^3 - 4x^2 + 7x - \frac{8}{3} & , \quad 1 \leq x < 3; \\ (x-2)\log_e(x-2) - x + \frac{10}{3} & , \quad x \geq 3 \end{cases}$$

Then which of the following options is/are correct

- (A) f is increasing on $(-\infty, 0)$ (B) f' has a local maximum at $x = 1$
 (C) f' is NOT differentiable at $x = 1$ (D) f is onto

Ans. D, A

Sol.

$$f(x) = \begin{cases} (x+1)^5 - 2x & x < 0 \\ x^2 - x + 1 & 0 \leq x < 1 \\ \frac{2}{3}x^3 - 4x^2 + 7x - \frac{8}{3} & 1 \leq x < 3 \\ (x-2)\ln(x-2) - x + \frac{10}{3} & x \geq 3 \end{cases}$$

$f(x)$ is continuous at $x = 0, x = 1, x = 3$

$\Rightarrow$ continuous for all $x \in \mathbb{R}$

$$f'(x) = \begin{cases} 5(x+1)^4 - 2 & x < 0 \\ 2x - 1 & 0 \leq x < 1 \\ 2x^3 - 8x + 7 & 1 < x < 3 \\ \ln(x-2) & x > 3 \end{cases}$$

$f'(x)$ is continuous at $x = 1$,

Discountinuous at $x = 0, 3$

Option (1) for $x < 0$ $f'(x)$ is changing its sign

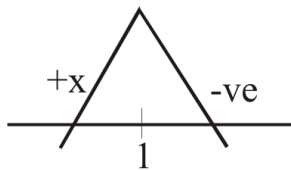
$\Rightarrow$ not increasing in $(0, \infty)$

Option (2) at $x = 1$

$$f''(x) = \begin{cases} 2 & 0 < x < 1 \\ 4x - 8 & 1 < x < 3 \end{cases}$$

$$f''(1^+) = -ve, f''(1^-) = +ve$$

f' has a local maxm at $x = 1$



$$\text{Option (3) } f'(1^+) = -4 \neq f'(1^-) = 2$$

f' is not differentiable at $x = 1$

Option (4) for $x \in \mathbb{R}$

$$\text{for } x < 0, (x+1)^5 - 2x \in (-\infty, 1)$$

$$\text{for } x \geq 3, (x-2)\ln(x-2) - x + \frac{10}{3} \in \left[\frac{1}{3}, \infty\right)$$

$$\text{Range of } f(x) = (-\infty, \infty)$$

Q27. Let $f(x) = \frac{\sin \pi x}{x^2}, x > 0$

Let $x_1 < x_2 < x_3 < \dots < x_n < \dots$ be all the points of local maximum of f and $y_1 < y_2 < y_3 < \dots < y_n < \dots$ be all the points of local minimum of f then which of the following options are correct

(A) $x_{n+1} - x_n > 2$ for every n (B) $x_1 < y_1$ (C) $x_n \in \left(2n, 2n + \frac{1}{2}\right)$ for every n

(D) $|x_n - y_n| > 1$ for every n

Ans. A, C, D

Sol. $f(x) = \frac{\sin \pi x}{x^2}, x > 0$

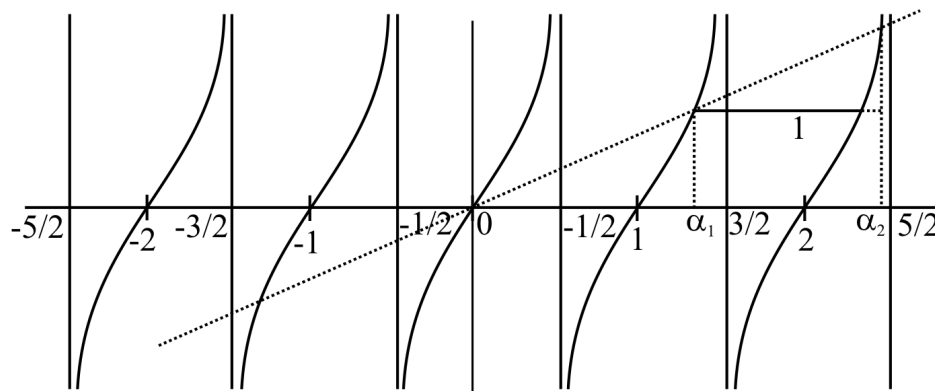
$$f'(x) = \frac{x^2 \cos(\pi x) \pi - \sin(\pi x) \cdot 2x}{x^4}$$

$$= \frac{[\pi x \cos(\pi x) - 2 \sin(\pi x)]}{x^3}$$

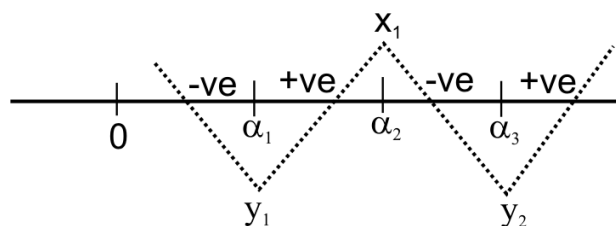
$$f'(x) = 0$$

$$\Rightarrow \pi x \cos \pi x = 2 \sin \pi x$$

$$\tan \pi x = \frac{\pi x}{2}$$



At 0^+ $f'(x) = -ve$



$$\alpha_1 = x_1$$

$$\alpha_2 = y_1 \text{ (and so on)}$$

$$\alpha_3 = x_2$$

$$x_1 > y_1$$

$$x_1 \in (2, 5/2)$$

using periodicity

$$x_n \in (2n, 2n+1/2)$$

$$x_{n+1} - x_n > 2 \quad \forall n \text{ correct}$$

$$|x_n - y_n| > 1 \quad \forall n \text{ correct}$$

Q28. Let $f_1 : (0, \infty) \rightarrow \mathbb{R}$ $f_2 : (0, \infty) \rightarrow \mathbb{R}$ be defined by

$$f_1(x) = \int_0^x \prod_{j=1}^{21} (t-j)^j dt, \quad x > 0$$

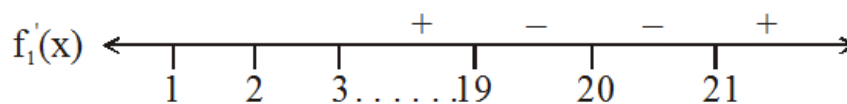
$$\text{and } f_2(x) = 98(x-1)^{50} - 600(x-1)^{49} + 2450, \quad x > 0,$$

where, for any positive integer n and real numbers $a_1, a_2, \dots, a_n$, $\prod_{i=1}^n a_i$ denotes the product of $a_1, a_2, \dots, a_n$. Let m_i and n_i , respectively, denote the number of points of local minima and the number of points of local maxima of function f_i , $i = 1, 2$, in the interval $(0, \infty)$.

The value of $2m_1 + 3n_1 + m_1n_1$ is _____.

Ans. 56.90, 57.10

Sol. $f_1(x) = \prod_{j=1}^{21} (x-j)^j = (x-j)^j = (x-2)^2 (x-3)^3 \dots (x-21)^{21}$



Points of minima $\Rightarrow x = 21, 17, 13, 9, 5, 1 \Rightarrow m_1 = 6$

Points of maxima $\Rightarrow x = 19, 15, 11, 7, 3 \Rightarrow n_1 = 5$

$$2m_1 + 3n_1 + m_1n_1 = 57.$$

$$f_2(x) = 4900(x-1)^{48}(x-7)$$

Points of minima $\Rightarrow x = 7$ $m_2 = 1$

$$n_2 = 0$$

$$6m_2 + 4n_2 + 8m_2n_2 = 6.$$

Q29. Let $f_1 : (0, \infty) \rightarrow \mathbb{R}$ $f_2 : (0, \infty) \rightarrow \mathbb{R}$ be defined by

$$f_1(x) = \int_0^x \prod_{j=1}^{21} (t-j)^j dt, \quad x > 0$$

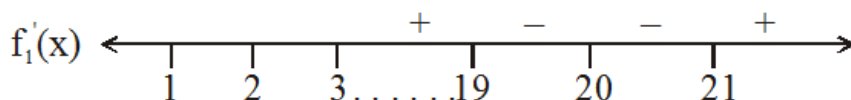
$$\text{and } f_2(x) = 98(x-1)^{50} - 600(x-1)^{49} + 2450, \quad x > 0,$$

where, for any positive integer n and real numbers $a_1, a_2, \dots, a_n$, $\prod_{i=1}^n a_i$ denotes the product of $a_1, a_2, \dots, a_n$. Let m_i and n_i , respectively, denote the number of points of local minima and the number of points of local maxima of function f_i , $i = 1, 2$, in the interval .

The value of $6m_2 + 4n_2 + 8m_2n_2$ is _____.

Ans. 5.90, 6.10

Sol. $f_1(x) = \prod_{j=1}^{21} (x-j) = (x-j)^j = (x-2)^2 (x-3)^3 \dots (x-21)^{21}$



Points of minima $\Rightarrow x = 21, 17, 13, 9, 5, 1 \Rightarrow m_1 = 6$

Points of maxima $\Rightarrow x = 19, 15, 11, 7, 3 \Rightarrow n_1 = 5$

$$2m_1 + 3n_1 + m_1n_1 = 57.$$

$$f_2(x) = 4900 (x-1)^{48} (x-7)$$

Points of minima $\Rightarrow x = 7$ $m_2 = 1$

$$n_2 = 0$$

$$6m_2 + 4n_2 + 8m_2n_2 = 6.$$

Q30. Let $\alpha = \sum_{k=1}^{\infty} \sin^{2k} \left(\frac{\pi}{6} \right)$. Let $g : [0, 1] \rightarrow \mathbb{R}$ be the function defined by $g(x) = 2^{\alpha x} + 2^{\alpha(1-x)}$

. Then, which of the following statements is/are TRUE ?

(A) The minimum value of $g(x)$ is $2^{7/6}$ (B) The maximum value of $g(x)$ is $1 + 2^{1/3}$

(C) The function $g(x)$ attains its maximum at more than one point

(D) The function $g(x)$ attains its minimum at more than one point

Ans. A, B, C

Sol. $\alpha = \sum_{k=1}^{\infty} \left(\frac{1}{2} \right)^{2k} = \sum_{k=1}^{\infty} \left(\frac{1}{4} \right)^k = \frac{1/4}{1 - 1/4} = \frac{1}{3}$

$$\text{Hence, } g(x) = 2^{x/3} + 2^{(1-x)/3}$$

$$\text{Now, } g'(x) = \frac{\ln 2}{3} \frac{(2^{2x/3} - 2^{1/3})}{2^{x/3}}$$

$$g'(x) = 0 \text{ at } x = \frac{1}{2}$$

And, derivative changes sign from negative to positive at $x = \frac{1}{2}$, hence $x = \frac{1}{2}$ is point of local minimum as well as absolute minimum of $g(x)$ for $x \in [0, 1]$.

Hence, minimum value of $g(x) = g\left(\frac{1}{2}\right)$

$$= 2^{1/6} + 2^{1/6} = 2^{7/6}$$

$\Rightarrow$ Option (A) is correct

Maximum value of $g(x)$ is either equal to $g(0)$ or $g(1)$.

$$g(0) = 1 + 2^{1/3}$$

$$g(1) = 2^{1/3} + 1$$

Hence (B) and (C) are also correct

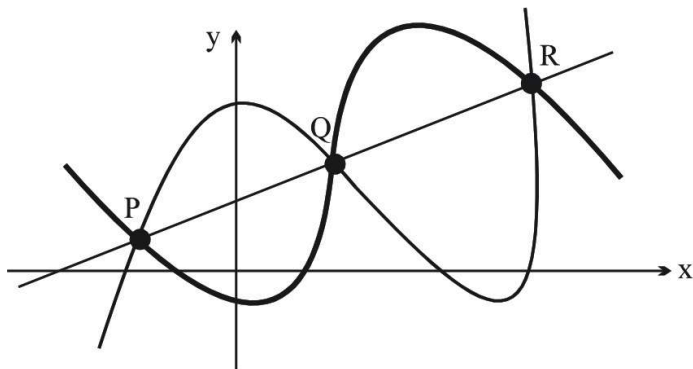
Exercise C1

Q1. The graphs $y = 2x^3 - 4x + 2$ and $y = x^3 + 2x - 1$ intersect in exactly 3 distinct points. The slope of the line passing through two of these points

(A) is equal to 4 (B) is equal to 6 (C) is equal to 8 (D) is not unique

Ans. C

Sol.



Let (x_1, y_1) and (x_2, y_2) are two of these points given $y = x^3 + 2x - 1$ and $y = 2x^3 - 4x + 2$

$$\therefore y_1 = 2x_1^3 - 4x_1 + 2 \quad \dots(1)$$

$$\text{and } 2y_1 = 2x_1^3 + 4x_1 - 2 \quad \dots(2)$$

$$(2) - (1)$$

$$y_1 = 8x_1 - 4 \quad \dots(3)$$

$$\text{|||ly } y_2 = 8x_2 - 4 \quad \dots(4)$$

$$y_2 - y_1 = 8(x_2 - x_1)$$

$$\frac{y_2 - y_1}{x_2 - x_1} = 8$$

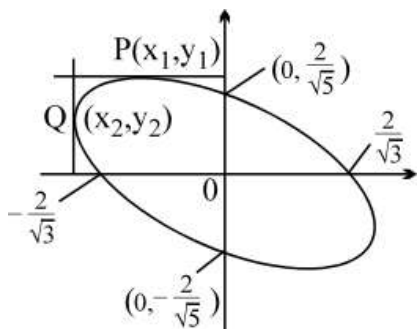
Q2. A function $y = f(x)$ has a second order derivative $f''(x) = 6(x - 1)$. If its graph passes through the point $(2, 1)$ and at that point the tangent to the graph is $y = 3x - 5$, then the function, is-

(A) $(x-1)^2$ (B) $(x-1)^3$ (C) $(x+1)^3$ (D) $(x+1)^2$

Ans. B

Sol. -

Q3. The lines $y = -\frac{3}{2}x$ and $y = -\frac{2}{5}x$ intersect the curve $3x^2 + 4xy + 5y^2 - 4 = 0$ at the points P and Q respectively. The tangents drawn to the curve at P and Q :



- (A) intersect each other at angle of 45° (B) are parallel to each other
 (C) are perpendicular to each other (D) none of these

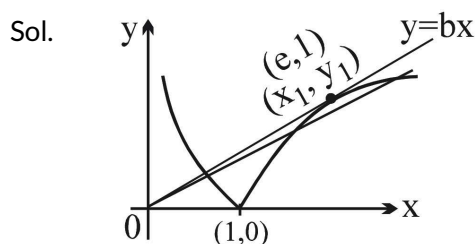
Ans. C

Sol. $\frac{dy}{dx} = -\left(\frac{2y+3x}{3x+5y}\right) \Rightarrow \frac{dy}{dx}\bigg|_{x_1y_1} = 0 \& \frac{dy}{dx}\bigg|_{x_2y_2} = \infty \Rightarrow$ tangents are perpendicular

Q4. The set of values of p for which the equation $|\ln x| - px = 0$ possess three distinct roots is

- (A) $\left(0, \frac{1}{e}\right)$ (B) $(0, 1)$ (C) $(1, e)$ (D) $(0, e)$

Ans. A



$$\frac{1}{x_1} = p = \frac{y_1}{x_1} \Rightarrow y_1 = 1 \& x_1 = e.$$

Hence slope of $y = px$ i.e. $p \in (0, 1/e)$ for three roots.

Q5. A curve is represented parametrically by the equations $x = t + e^{at}$ and $y = -t + e^{at}$ when $t \in \mathbb{R}$ and $a > 0$. If the curve touches the axis of x at the point A, then the coordinates of the point A are

- (A) $(1, 0)$ (B) $(1/e, 0)$ (C) $(e, 0)$ (D) $(2e, 0)$

Ans. D

Sol. -

Q6. A horse runs along a circle with a speed of 20 km/hr. A lantern is at the centre of the circle. A fence is along the tangent to the circle at the point at which the horse starts. The speed with which the shadow of the horse moves along the fence at the moment when it covers $1/8$ of the circle in km/hr is

- (A) 20 (B) 40 (C) 30 (D) 60

Ans. B

Sol. $\tan \theta = \frac{\pi}{r} \Rightarrow x = r \tan \theta$

$$\Rightarrow \frac{dx}{dt} = r \sec^2 \theta \left(\frac{d\theta}{dt} \right) = r \omega \sec^2 \theta = v \sec^2 \theta$$

where $\theta = \frac{\pi}{8}$, $\frac{dx}{dt} = v \sec^2 \left(\frac{\pi}{8} \right) = 2v = 40 \text{ km/hr}$; $\theta = 45^\circ$

Q7. If $f(x) = 4x^3 - x^2 - 2x + 1$ and $g(x) = \begin{cases} \min\{f(t) : 0 \leq t \leq x\} & ; 0 \leq x \leq 1 \\ 3 - x & ; 1 < x \leq 2 \end{cases}$ then

$g\left(\frac{1}{4}\right) + g\left(\frac{1}{4}\right) + g\left(\frac{1}{4}\right)$ has the value equal to :

- (A) $\frac{7}{4}$ (B) $\frac{9}{4}$ (C) $\frac{13}{4}$ (D) $\frac{5}{2}$

Ans. D

Sol. -

Q8. Let $f(x) = x^3 - 3x^2 + 2x$. If the equation $f(x) = k$ has exactly one positive and one negative solution then the value of k equals

- (A) $-\frac{2\sqrt{3}}{9}$ (B) $-\frac{2}{9}$ (C) $\frac{2}{3\sqrt{3}}$ (D) $\frac{1}{3\sqrt{3}}$

Ans. A

Sol. -

Q9. Let $f(x)$ and $g(x)$ be two continuous functions defined from $\mathbb{R} \rightarrow \mathbb{R}$, such that $f(x_1) > f(x_2)$ and

$g(x_1) < g(x_2)$, $\forall x_1 > x_2$, then solution set of $f(g(\alpha^2 - 2\alpha)) > f(g(3\alpha - 4))$ is

- (A) $\mathbb{R}$ (B) ϕ (C) $(1, 4)$ (D) $\mathbb{R} - [1, 4]$

Ans. C

Sol. -

Q10. Given $f'(1) = 1$ and $\frac{d}{dx}(f(2x)) = f'(x) \forall x > 0$. If $f'(x)$ is differentiable then there exists a number $c \in (2, 4)$ such that $f''(c)$ equals

- (A) $-1/4$ (B) $-1/8$ (C) $1/4$ (D) $1/8$

Ans. B

Sol. -

Q11. If $f(x) = \frac{x^2}{2 - 2\cos x}$; $g(x) = \frac{x^2}{6x - 6\sin x}$ where $0 < x < 1$, then

- (A) both 'f' and 'g' are increasing functions
- (B) 'f' is decreasing & 'g' is increasing function
- (C) 'f' is increasing & 'g' is decreasing function
- (D) both 'f' & 'g' are decreasing function

Ans. C

Sol. -

Q12. Let $f(x) = \begin{cases} x^\alpha \ln x, & x > 0 \\ 0, & x = 0 \end{cases}$. Rolle's theorem is applicable to f for $x \in [0, 1]$, if $\alpha =$

- (A) -2
- (B) -1
- (C) 0
- (D) 1/2

Ans. D

Sol. -

Q13. If f is a strictly increasing function, then $\lim_{x \rightarrow 0} \frac{f(x^2) - f(x)}{f(x) - f(0)}$ is equal to

- (A) 0
- (B) 1
- (C) -1
- (D) 2

Ans. C

Sol. -

Q14. Given that $f(x)$ is continuously differentiable on $a \leq x \leq b$ where $a < b$, $f(a) < 0$ and $f(b) > 0$, which of the following are always true?

- (i) $f(x)$ is bounded on $a \leq x \leq b$.
- (ii) The equation $f(x) = 0$ has at least one solution in $a < x < b$.
- (iii) The maximum and minimum values of $f(x)$ on $a \leq x \leq b$ occur at points where $f'(c) = 0$.
- (iv) There is at least one point c with $a < c < b$ where $f'(c) > 0$.
- (v) There is at least one point d with $a < d < b$ where $f'(c) < 0$.

- (A) only (ii) and (iv) are true
- (B) all but (iii) are true
- (C) all but (v) are true
- (D) only (i), (ii) and (iv) are true

Ans. D

Sol. (i) This statement is true, every continuous function is bounded on a closed interval
(ii) True again, by Intermediate Value Theorem
(iii) Not true, because maximum and / or minimum values could also occur at a or b , without the derivatives being 0.
(iv) True. By the Mean Value Theorem there exist a point between a and b where the derivative is exactly $\frac{f(b) - f(a)}{b - a}$, a clearly positive value.

(v) Not always true, for example the function might be strictly increasing guaranteeing the derivative to be always positive.

Thus the true statements are (i), (ii) and (iv) and the correct answer is (D).

Q15. Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false.

Statement-1: If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $c \in \mathbb{R}$ is such that f is increasing in $(c - \delta, c)$ and f is decreasing in

$(c, c + \delta)$ then f has a local maximum at c . Where δ is a sufficiently small positive quantity.

Statement-2 : Let $f : (a, b) \rightarrow \mathbb{R}$, $c \in (a, b)$. Then f can not have both a local maximum and a point of inflection at $x = c$.

Statement-3 : The function $f(x) = x^2 |x|$ is twice differentiable at $x = 0$.

Statement-4 : Let $f : [c - 1, c + 1] \rightarrow [a, b]$ be bijective map such that f is differentiable at c then f^{-1} is also differentiable at $f(c)$.

(A) FFTF (B) TTFT (C) FTTF (D) TTTF

Ans. C

Sol. -

Q16. In a regular triangular prism the distance from the centre of one base to one of the vertices of the other base is l . The altitude of the prism for which the volume is greatest :

(A) $\frac{l}{2}$ (B) $\frac{l}{\sqrt{3}}$ (C) $\frac{l}{3}$ (D) $\frac{l}{4}$

Ans. B

Sol. -

Q17. Let h be a twice continuously differentiable positive function on an open interval J .

Let

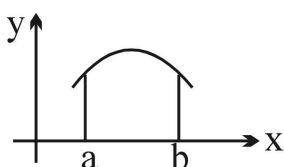
$g(x) = \ln$ for each $x \in J$

Suppose $(h'(x))^2 > h''(x)h(x)$ for each $x \in J$. Then

(A) g is increasing on J (B) g is decreasing on J (C) g is concave up on J
(D) g is concave down on J

Ans. D

Sol.



Given $g(x) = \ln(h(x))$

$$g'(x) = \frac{h'(x)}{h(x)}$$

$$g''(x) = \frac{h'(x)h''(x) - h'(x)^2}{h^2(x)} < 0 \quad (\text{given})$$

$\therefore g''(x) < 0 \Rightarrow g(x)$ is concave down.

Q18. If the function $f(x) = \frac{t + 3x - x^2}{x - 4}$, where 't' is a parameter has a minimum and a maximum then the range of values of 't' is

- (A) $(0, 4)$ (B) $(0, \infty)$ (C) $(-\infty, 4)$ (D) $(4, \infty)$

Ans. C

Sol. -

Q19. P and Q are two points on a circle of centre C and radius a, the angle PCQ being 2θ then the radius of the circle inscribed in the triangle CPQ is maximum when

- (A) $\sin \theta = \frac{\sqrt{3} - 1}{2\sqrt{2}}$ (B) $\sin \theta = \frac{\sqrt{5} - 1}{2}$ (C) $\sin \theta = \frac{\sqrt{5} + 1}{2}$ (D) $\sin \theta = \frac{\sqrt{5} - 1}{4}$

Ans. B

Sol. -

Q20. The range of values of m for which the line $y = mx$ and the curve $y = \frac{x}{x^2 + 1}$ enclose a region, is

- (A) $(-1, 1)$ (B) $(0, 1)$ (C) $[0, 1]$ (D) $(1, \infty)$

Ans. B

Sol. -

Q21. For a steamer the consumption of petrol (per hour) varies as the cube of its speed (in km). If the speed of the current is steady at C km/hr then the most economical speed of the steamer going against the current will be

- (A) $1.25 C$ (B) $1.5 C$ (C) $1.75 C$ (D) $2 C$

Ans. B

Sol. Time of journey = $\frac{d}{V - C}$ when V is the speed in still water.

$\therefore$ petrol burnt per hour = kV^3

$$\therefore \text{fuel cost} = \frac{kV^3 d}{V - C} = kd = \frac{V^3}{V - C} = f(v)$$

now proceed.]

Q22. Let f , g and h be real-valued functions defined on the interval $[0, 1]$ by $f(x) = e^{x^2} + e^{-x^2}$, $g(x) = xe^{x^2} + e^{-x^2}$ and $h(x) = x^2e^{x^2} + e^{-x^2}$. If a , b and c denote respectively, the absolute maximum of f , g and h on $[0, 1]$, then

- (A) $a = b$ and $c \neq b$ (B) $a = c$ and $a \neq b$ (C) $a \neq b$ and $c \neq b$ (D) $a = b = c$

Ans. D

Sol. If $x \in [0, 1]$ then $x^2 \leq x \leq 1$

$$x^2e^{x^2} \leq xe^{x^2} \leq e^{x^2}$$

Add e^{-x^2} to all sides

$$\Rightarrow h(x) \leq g(x) \leq f(x) \quad \dots\dots\dots (i)$$

where, $f(x) = e^{x^2} + e^{-x^2}$

$$f'(x) = 2x(e^{x^2} - e^{-x^2}) > 0$$

$\Rightarrow f(x)$ has a maxima at $x = 1$

$$\Rightarrow a = e + \frac{1}{e}$$

$$h(x) = x^2e^{x^2} + e^{-x^2}$$

$$h'(x) = 2x^3e^{x^2} + 2xe^{x^2} - 2xe^{-x^2}$$

$$= 2x^3e^{x^2} + 2x(e^{x^2} - e^{-x^2}) > 0$$

$\Rightarrow h(x)$ has a maxima at $x = 1$

$$\Rightarrow c = e + \frac{1}{e}$$

$$\therefore h(x) \leq f(x)$$

$\Rightarrow g(x)$ also has a maximum value at $x = 1$

$$\Rightarrow a = b = c$$

Q23. Consider the curve $f(x) = x^{1/3}$, then

- (A) the equation of tangent at $(0, 0)$ is $x = 0$
(B) the equation of normal at $(0, 0)$ is $y = 0$
(C) normal to the curve does not exist at $(0, 0)$
(D) $f(x)$ and its inverse meet at exactly 3 points.

Ans. D, B, A

Sol. -

Q24. If the line, $ax + by + c = 0$ is a normal to the curve $xy = 2$, then:

- (A) $a < 0$, $b > 0$ (B) $a > 0$, $b < 0$ (C) $a > 0$, $b > 0$ (D) $a < 0$, $b < 0$

Ans. A, B

Sol. Slope of the normal $(ax + by + c = 0) = -\frac{a}{b}$

The given curve is

$$xy = 2$$

$$\Rightarrow \frac{dy}{dx} = -\frac{y}{x} = \frac{b}{a} \Rightarrow \frac{b}{a} < 0$$

$$\Rightarrow a < 0, b > 0 \text{ or } a > 0, b < 0$$

Q25. The normal to the curve $x = a(\cos \theta + \theta \sin \theta)$, $y = a(\sin \theta - \theta \cos \theta)$ at any point ' θ ' is such that -

(A) it passes through the origin (B) it makes an angle $\frac{\pi}{2} + \theta$ with the x-axis

(C) it passes through $\left(a\frac{\pi}{2}, -a\right)$ (D) it is at a constant distance from the origin.

Ans. D, B

Sol. -

Q26. Which of the following functions do not satisfy conditions of Rolle's Theorem ?

(A) $e^x \sin x$, $x \in \left[0, \frac{\pi}{2}\right]$ (B) $(x+1)^2(2x-3)^5$, $x \in \left[-1, \frac{3}{2}\right]$ (C) $\sin |x|$, $x \in [\pi, 2\pi]$

(D) $\sin \frac{1}{x}$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Ans. D, A

Sol. -

Q27. The function $y = 2x^2 - \ln |x|$ is monotonically increasing in the interval I_1 and monotonically decreasing in the interval I_2 , $x (\neq 0)$, then

(A) $I_1 = \left(-\frac{1}{2}, 0\right) \cup \left(\frac{1}{2}, \infty\right)$ (B) $I_2 = \left(-\infty, -\frac{1}{2}\right) \cup \left(0, \frac{1}{2}\right)$

(C) $I_1 = \left(-\infty, -\frac{1}{2}\right) \cup \left(0, \frac{1}{2}\right)$ (D) $I_2 = \left(-\frac{1}{2}, 0\right) \cup \left(\frac{1}{2}, \infty\right)$

Ans. A, B

Sol. -

Q28. For the function $f(x) = x^4(12 \ln x - 7)$

(A) the point $(1, -7)$ is the point of inflection (B) $x = e^{1/3}$ is the point of minima

(C) the graph is concave downwards in $(0, 1)$

(D) the graph is concave upwards in $(1, \infty)$

Ans. A, B, C, D

Sol. -

Q29. Which of the following statements is/are True?

- (A) If f has domain $[0, \infty)$ and has no horizontal asymptotes then $\lim_{x \rightarrow \infty} f(x) = \infty$ or $\lim_{x \rightarrow \infty} f(x) = -\infty$.
- (B) If f is continuous on $[-1, 1]$ and $f(-1) = 4$ and $f(1) = 3$ then there exist a number r such that $|r| < 1$ and $f(r) = \pi$.
- (C) $\lim_{x \rightarrow \infty} \left(\frac{x+1}{x} \right)$ does not exist.
- (D) For all values of $m \in \mathbb{R}$ the line $y - mx + m - 1 = 0$ cuts the circle $x^2 + y^2 - 2x - 2y + 1 = 0$ orthogonally.

Ans. D, C, B

Sol. -

Q30. A function f is defined by $f(x) = \int_0^{\pi} \cos t \cos(x-t) dt$, $0 \leq x \leq 2\pi$ then which of the following hold(s) good?

- (A) $f(x)$ is continuous but not differentiable in $(0, 2\pi)$ (B) Maximum value of f is π
- (C) There exists atleast one $c \in (0, 2\pi)$ s.t. $f'(c) = 0$. (D) Minimum value of f is $-\frac{\pi}{2}$.

Ans. C, D

Sol. -

Q31. Let $f(x)$ be a polynomial of degree 3 having a local maximum at $x = -1$. If $f(-1) = 2$, $f(3) = 18$, and $f'(x)$ has a local minimum at $x = 0$, then

- (A) distance between $(-1, 2)$ and $(a, f(a))$, which are the points of local maximum and local minimum on the curve $y = f(x)$ is $2\sqrt{5}$
- (B) $f(x)$ is a decreasing function for $1 \leq x \leq 2\sqrt{5}$
- (C) $f'(x)$ has a local maximum at $x = 2\sqrt{5}$ (D) $f(x)$ has a local minimum at $x = 1$

Ans. D, A

Sol. -

Q32. Consider the function $f(x) = x^2 - x \sin x - \cos x$ then the statements which holds good, are

- (A) $f(x) = k$ has no solution for $k < -1$.

(B) f is increasing for $x < 0$ and decreasing for $x > 0$. (C) $\lim_{x \rightarrow \pm\infty} f(x) \rightarrow \infty$

(D) The zeros of $f(x) = 0$ lie on the same side of the origin.

Ans. A, C

Sol. -

Q33. For the function $f(x) = \ln(1 - \ln x)$ which of the following do not hold good?

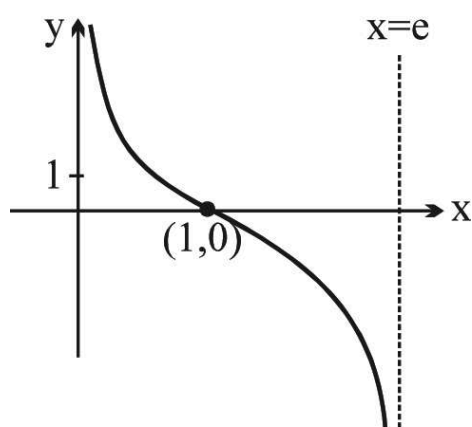
(A) increasing in $(0, 1)$ and decreasing in $(1, e)$

(B) decreasing in $(0, 1)$ and increasing in $(1, e)$

(C) $x = 1$ is the critical number for $f(x)$. (D) f has two asymptotes

Ans. A, B, C

Sol.



$$f(x) = \ln(1 - \ln x)$$

domain $(0, e)$

$$f'(x) = \frac{1}{(1 - \ln x)} \cdot \frac{1}{x} < 0 \Rightarrow \text{decreasing } \forall x \text{ in its domain } \quad \forall (A) \& (B) \text{ are incorrect}$$

$$f'(1) = -1 \Rightarrow (C) \text{ is also incorrect}$$

$$\text{also } f(1) = 0; \quad \lim_{x \rightarrow e^-} f(x) \rightarrow \infty; \quad \lim_{x \rightarrow 0^+} f(x) \rightarrow \infty$$

$$f''(x) = \frac{-\ln x}{x^2(1 - \ln x)^2}$$

$f''(1) = 0$ which is a point of inflection graph is as shown y axis and $x = e$ are two asymptotes

Q34. Which of the following statement(s) is/are True?

If f is a continuous real valued function defined on $[0, 1]$ then \$ positive number c

(A) such that

$$|f(x) - f(y)| \leq c \text{ for all } x, y \text{ in } [0, 1].$$

- (B) If f is continuous on $\mathbb{R}$ such that $\lim_{x \rightarrow -\infty} f(x) = -\infty$ and $\lim_{x \rightarrow \infty} f(x) = \infty$ then the equation $f(x) = k$, $k \in \mathbb{R}$ has at least a root in $\mathbb{R}$.
- (C) If f and g are derivable and $f(x) > g(x)$ for $a < x < b$ then $f'(x) > g'(x)$ for $a < x < b$.
- (D) If f and g are continuous and $f(x) > g(x)$ for $a < x < b$ then $\int_a^b f(x) dx > \int_a^b g(x) dx$.

Ans. A, B, D

Sol. -